

Rank-Recursive Iterated Extended Kalman Filtering: Exact Gaussian Updates from Low-Rank Multisecant Relinearization

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Abstract—The iterated extended Kalman filter (IEKF) repeatedly relinearizes a nonlinear measurement model. Conventional implementations typically rebuild and refactor the associated Gaussian update at every inner iteration, even when the new linearization is represented by a low-rank quasi-Newton correction. This paper develops a rank-recursive formulation that propagates the *change* in the linearization rather than recomputing the entire update. A rank- q multisecant correction of the measurement Jacobian is shown to induce rank-at-most- $2q$ modifications of both the innovation and posterior-information matrices. Consequently, the innovation inverse, posterior covariance, Kalman gain, log-determinant, normalized innovation squared, and Gaussian likelihood can be updated algebraically exactly through a dense core of size at most $2q \times 2q$. A prior-metric least-change multisecant specialization is also derived. Its closed form depends on the prior covariance but, under hard secant constraints, the left measurement metric cancels; this distinction is made explicit. The resulting update is coordinate-covariant and minimizes a direct upper bound on normalized innovation-covariance perturbation, but is not claimed to dominate Euclidean Broyden in nonlinear convergence. In 3000 anisotropic secant stress tests it reduces the median innovation-covariance perturbation by $7.29\times$, $38.5\times$, and $282.8\times$ at prior condition numbers 10^2 , 10^3 , and 10^4 , respectively. Rank-recursive updates with $q \in \{1, 2, 4, 8\}$ reproduce dense recomputation to 10^{-12} – 10^{-15} relative accuracy. A nonlinear AC IEEE 39-bus experiment with 77 states and 108 measurements then evaluates the complete safeguarded algorithm: rank-recursive quasi-Newton variants require only about 1.0–1.25 analytic Jacobian evaluations per update versus 4.0–4.5 for the exact IEKF, while the Gaussian recursion remains exact to numerical precision. The exact Jacobian changes are not themselves strongly low-rank, clarifying that the method is a controlled low-rank curvature surrogate rather than a truncation of the exact Jacobian. A square-root FPGA mapping is specified, while synthesizable VHDL and post-route measurements are deliberately left for subsequent work.

Index Terms—iterated extended Kalman filter, quasi-Newton methods, multisecant methods, low-rank updates, Woodbury identity, square-root filtering, nonlinear state estimation, FPGA mapping, power-system state estimation.

I. INTRODUCTION

THE extended Kalman filter (EKF) remains a standard nonlinear recursive estimator because it combines a probabilistic state representation with local linearization. The iterated EKF (IEKF) improves the measurement update by relinearizing the observation model around successive posterior iterates and can be interpreted as Gauss–Newton optimization of a local maximum-a-posteriori (MAP) objective [1]. This additional robustness comes with a repeated linear-algebra cost: a new Jacobian changes the innovation covariance, factorization,

gain, and posterior covariance. Treating every inner iteration as a new dense problem discards substantial algebraic reuse.

This paper asks a narrower question than approximate-inverse or large-scale Kalman filtering: *if a quasi-Newton IEKF deliberately changes its Jacobian model by rank q , how much of the Gaussian update truly changes?* The answer is that the relevant covariance and information matrices change by rank at most $2q$. Once this structure is exposed, repeated dense refactorization is unnecessary: the Gaussian algebra associated with the selected low-rank linearization can be propagated exactly using a small $2q \times 2q$ core.

The distinction between *nonlinear-model approximation* and *linear-algebra approximation* is central. Broyden and multisecant methods provide low-rank surrogate curvature. They do not generally equal the exact Jacobian. The contribution here is not to hide that approximation, but to prevent a second, unnecessary approximation or refactorization cost in the Gaussian step. Conditioned on the selected rank- q Jacobian correction, the subsequent innovation/information recursion is algebraically exact.

A second question is how a secant correction should be measured. Standard good Broyden minimizes an unweighted Frobenius change. The estimator supplies a physically meaningful state metric through the prior covariance P^- . We therefore derive a prior-metric least-change multisecant rule. An important correction relative to an overly broad “prior-and-noise metric” interpretation is that, under exact hard secant constraints, any nonsingular left measurement weighting cancels from the closed-form solution. The update is thus genuinely *prior-metric*; measurement whitening remains useful for the Gaussian recursion and numerical conditioning, not for changing the hard-constrained secant solution.

The main contributions are:

- A prior-metric multisecant specialization of least-change quasi-Newton relinearization, together with an explicit statement of the measurement-metric cancellation and affine coordinate covariance.
- A rank-closure result showing that a rank- q Jacobian correction induces rank-at-most- $2q$ changes in the innovation matrix, information matrix, posterior covariance, and—more tightly than a naive sum-of-ranks argument—the Kalman gain.
- Exact $2q$ -core recursions for the innovation/information inverse, posterior covariance, gain, log-determinant, NIS, and likelihood, including explicit low-dimensional formulas useful for implementation.

- A safeguarded rank-recursive IEKF that uses MAP descent and secant conditioning for globalization. NIS is retained as a post-update diagnostic, not used as an unjustified inner-loop χ^2 test on repeatedly reused data.
- An end-to-end nonlinear AC IEEE 39-bus study in which the quasi-Newton IEKF actually propagates the innovation inverse recursively rather than recomputing it. Evaluation uses posterior Mahalanobis distance, MAP objective gap, physical state RMSE, refresh rate, and Jacobian-evaluation count.
- A hardware mapping and VHDL validation plan. No RTL, timing, LUT/DSP/BRAM, or power claims are made in this paper.

II. RELATED WORK AND SCOPE

A. IEKF and Quasi-Newton Filtering

Bell and Cathey established the Gauss–Newton interpretation of the IEKF [1]. Quasi-Newton machinery has subsequently appeared in several filtering contexts. Auvinen *et al.* used limited-memory BFGS for large-scale KF/EKF calculations [2]. Mukherjee *et al.* used Broyden rank-one updates to avoid repeatedly evaluating process and measurement Jacobians [3]. Kullberg *et al.* explicitly analyzed the relationship between quasi-Newton IEKF formulations and iterated statistical-linearization filters [4]. Accordingly, this paper does *not* claim that quasi-Newton filtering or weighted secant updates are new. The claimed contribution is the low-rank closure of the Gaussian update induced by such relinearization and its exact recursive propagation.

Least-change secant methods and weighted secant geometries are classical numerical-analysis tools [5]–[7]. Our prior-metric formula is best viewed as a filtering-specific specialization whose estimator geometry provides the state weighting and whose low-rank factorization feeds the rank-recursive Gaussian algebra.

B. Incremental Factorization and Fast Kalman Recursions

iSAM updates sparse square-root information factorizations incrementally instead of rebuilding them from scratch, and iSAM2 extends this idea to nonlinear incremental smoothing with fluid relinearization [8], [9]. These methods provide important conceptual prior art for “recompute only what changed.” Their setting, however, is sparse factor-graph smoothing rather than the dense within-measurement IEKF recursion studied here. Chandrasekhar-type Kalman algorithms similarly propagate low-displacement-rank covariance differences across filtering time; robust hyperbolic/SVD realizations have been developed for that recursion [10], [11]. Our rank event is different: it is generated by a low-rank *relinearization* within a nonlinear measurement update.

C. Approximate, Matrix-Free, and Hardware Kalman Filters

Approximate and inverse-free Kalman filters reduce dense solve cost through Taylor, iterative, or Krylov approximations [12]–[14]. Computation-aware Kalman filtering explicitly models numerical approximation uncertainty [15]. KalmMind

exposes tunable accuracy/cost policies in FPGA-oriented Kalman accelerators [16], [17]. Those methods trade algebraic accuracy or representation fidelity for computation. The present method instead keeps the Gaussian algebra exact for the chosen low-rank Jacobian surrogate; its approximation is localized in the nonlinear curvature model.

III. IEKF GEOMETRY AND PRIOR-METRIC MULTISECANTS

A. Whitened IEKF

Consider a prior $x \sim \mathcal{N}(x^-, P^-)$ and nonlinear measurement

$$y = h(x) + v, \quad v \sim \mathcal{N}(0, R), \quad (1)$$

with $P^- \succ 0$ and $R \succ 0$. Write square roots $P^- = LL^T$ and $R = MM^T$, and define normalized coordinates

$$\xi = L^{-1}(x - x^-), \quad \tilde{y} = M^{-1}y. \quad (2)$$

The local MAP objective becomes

$$\Phi(\xi) = \frac{1}{2} \|\xi\|^2 + \frac{1}{2} \left\| \tilde{y} - \tilde{h}(\xi) \right\|^2, \quad (3)$$

where the whitened Jacobian is

$$G = M^{-1}HL. \quad (4)$$

The two dual SPD matrices are

$$\bar{S} = I + GG^T, \quad \bar{\Lambda} = I + G^T G. \quad (5)$$

A square-root implementation can maintain L directly, so prior whitening need not introduce a new $\mathcal{O}(n^3)$ factorization at each measurement update; if L is not already available, its cost must be counted explicitly.

B. The IEKF Inner Problem and the Recentered Residual

At inner iteration j , let $H_j = \partial h / \partial x|_{x_j}$. The standard recentered IEKF measurement residual is

$$\nu_j = y - h(x_j) + H_j(x_j - x^-), \quad (6)$$

and the corresponding Gaussian subproblem gives

$$x_{j+1}^{\text{GN}} = x^- + P^- H_j^T (H_j P^- H_j^T + R)^{-1} \nu_j. \quad (7)$$

Equivalently, in normalized state coordinates, the Gauss–Newton normal matrix is $I + G_j^T G_j$ while the dual innovation matrix is $I + G_j G_j^T$. The rank-recursive method changes neither (6) nor the target of this local Gaussian subproblem. It changes only how the matrices associated with H_{j+1} are obtained from those associated with H_j .

This distinction is useful for interpreting “exactness.” If a quasi-Newton rule proposes $\hat{H}_{j+1}$, then the recursion in Section IV returns the same Gaussian mean/covariance that dense IEKF algebra would return for $\hat{H}_{j+1}$, to floating-point accuracy. It does *not* assert that $\hat{H}_{j+1}$ equals the analytic Jacobian. Consequently, comparisons to an exact-Jacobian IEKF measure nonlinear-model error, whereas comparisons between dense and rank-recursive algebra for the same $\hat{H}_{j+1}$ measure implementation error. The experiments report both separately.

TABLE I
POSITIONING RELATIVE TO DIRECTLY RELEVANT PRIOR WORK.

Method	Reuse mechanism	Nonlinear model	Linear algebra	Distinction from this work
Broyden-EKF [3]	Rank-one Jacobian update	Approximate Jacobian	Conventional EKF algebra	Does not propagate the induced rank- $2q$ Gaussian change
QN-IEKF [4]	Quasi-Newton Hessian/linearization view	Quasi-Newton	Standard iterated filtering	Establishes QN filtering relationships, not rank-closure recursion
LBFGS-KF/EKF [2]	Limited-memory inverse-Hessian structure	Optimization surrogate	Large-scale iterative computation	Different large-scale optimization formulation
iSAM/iSAM2 [8], [9]	Incremental sparse factorization	Selective relinearization	Exact sparse QR/Bayes-tree update	Smoothing/factor-graph setting; not dense IEKF rank event
Chandrasekhar [10], [11]	Low-rank covariance differences across time	Linear model	Exact fast KF recursion	Exploits Riccati displacement rank, not relinearization rank
KalmMind [16], [17]	Temporal reuse, configurable approximation	Fixed KF model	Tunable inverse approximation	Trades inverse accuracy for hardware cost
This work	Rank-q multisecant relinearization	Controlled low-rank surrogate	Exact rank-$2q$ Gaussian recursion	Within-IEKF change propagation; VHDL mapping left future

C. Consistent Multisecant Definition

At a current linearization H , collect q state secants and measurement differences

$$X = [s_1, \dots, s_q], \quad Y = [\Delta h_1, \dots, \Delta h_q], \quad (8)$$

all referenced to the same current Jacobian model. A correction ΔH should satisfy

$$(H + \Delta H)X = Y, \quad (9)$$

or equivalently

$$\Delta H X = E, \quad E := Y - H X. \quad (10)$$

This definition avoids mixing secant defects formed with different historical Jacobians.

D. Prior-Metric Least-Change Update

Consider the weighted least-change problem

$$\min_{\Delta H} \left\| R^{-1/2} \Delta H (P^-)^{1/2} \right\|_F^2 \quad \text{s.t.} \quad \Delta H X = E. \quad (11)$$

Theorem 1 (prior-metric multisecant). If $X^T (P^-)^{-1} X \succ 0$, the unique minimum-norm solution of (11) is

$$\Delta H_P = E (X^T (P^-)^{-1} X)^{-1} X^T (P^-)^{-1}. \quad (12)$$

It has rank at most q .

Proof. Let $Z = R^{-1/2} \Delta H (P^-)^{1/2}$, $A = (P^-)^{-1/2} X$, and $B = R^{-1/2} E$. Then $ZA = B$. The minimum-Frobenius-norm solution is $Z^* = BA^\dagger$. Transforming back yields (12). The left factor $R^{-1/2}$ cancels because the secant constraint is hard and exact; the resulting optimizer therefore depends on the prior metric but not on R . The rank bound follows from the factorization through $E \in \mathbb{R}^{m \times q}$. $\square$

For $q = 1$,

$$\Delta H_P = \frac{e s^T (P^-)^{-1}}{s^T (P^-)^{-1} s}, \quad (13)$$

which reduces to good Broyden when $P^- = cI$.

Proposition 1 (linear-coordinate covariance). Under $x = Tz$ with nonsingular T , the update transforms as

$\Delta H_z = \Delta H_x T$. Hence arbitrary linear state rescaling does not change the modeled secant geometry when the covariance is transformed consistently.

E. Why the Prior Metric Is Relevant, but Not Universally Faster

Let

$$Z = R^{-1/2} \Delta H (P^-)^{1/2}, \quad G = R^{-1/2} H (P^-)^{1/2}. \quad (14)$$

The normalized innovation matrix changes by

$$\Delta \bar{S} = G Z^T + Z G^T + Z Z^T, \quad (15)$$

and therefore

$$\|\Delta \bar{S}\|_F \leq 2 \|G\|_2 \|Z\|_F + \|Z\|_F^2. \quad (16)$$

Theorem 1 minimizes $\|Z\|_F$ among exact multisecant corrections, hence also minimizes the right-hand side of (16). This is a geometric guarantee, not a global nonlinear-convergence theorem. Section VII shows explicitly that Euclidean and prior-metric multisecants can each be better in different nonlinear regimes.

IV. RANK-RECURSIVE GAUSSIAN ALGEBRA

Assume a selected whitened Jacobian surrogate changes as

$$G^+ = G + UV^T, \quad (17)$$

where $U \in \mathbb{R}^{m \times q}$ and $V \in \mathbb{R}^{n \times q}$. The correction may come from (12), ordinary multisecant Broyden, or another rank- q model.

A. Measurement-Space Closure

From (5),

$$\bar{S}^+ - \bar{S} = G V U^T + U V^T G^T + U (V^T V) U^T. \quad (18)$$

Define

$$A = GV, \quad D = V^T V, \quad B_S = [A \ U], \quad J_S = \begin{bmatrix} 0 & I_q \\ I_q & D \end{bmatrix}. \quad (19)$$

TABLE II
RANK CLOSURE INDUCED BY A RANK- q RELINEARIZATION.

Quantity	Rank of change
Jacobian surrogate G	$\leq q$
Innovation matrix $I + GG^T$	$\leq 2q$
Information matrix $I + G^T G$	$\leq 2q$
Posterior covariance/information inverse	$\leq 2q$
Kalman gain	$\leq 2q$
New dense core	$2q \times 2q$

Then

$$\bar{S}^+ = \bar{S} + B_S J_S B_S^T, \quad \text{rank}(\Delta \bar{S}) \leq 2q. \quad (20)$$

Moreover,

$$J_S^{-1} = \begin{bmatrix} -D & I_q \\ I_q & 0 \end{bmatrix}. \quad (21)$$

With $W = \bar{S}^{-1}$ and

$$M_S = J_S^{-1} + B_S^T W B_S, \quad (22)$$

Woodbury gives the exact recursion

$$W^+ = W - W B_S M_S^{-1} B_S^T W. \quad (23)$$

The newly factorized dense matrix is only $2q \times 2q$.

B. Information-Space Closure

Similarly, with

$$C = G^T U, \quad F = U^T U, \quad B_\Lambda = [C \ V], \quad J_\Lambda = \begin{bmatrix} 0 & I_q \\ I_q & F \end{bmatrix}, \quad (24)$$

we obtain

$$\bar{\Lambda}^+ = \bar{\Lambda} + B_\Lambda J_\Lambda B_\Lambda^T, \quad \text{rank}(\Delta \bar{\Lambda}) \leq 2q. \quad (25)$$

Thus a dense implementation can select measurement or information coordinates according to dimension, sparsity, and factor availability rather than always forming the larger system.

Theorem 2 (rank closure including the gain). Let $\bar{P} = \bar{\Lambda}^{-1}$ and $\bar{K} = \bar{P} G^T$. Under (17),

$$\text{rank}(\Delta \bar{S}), \text{rank}(\Delta \bar{\Lambda}), \text{rank}(\Delta \bar{P}), \text{rank}(\Delta \bar{K}) \leq 2q. \quad (26)$$

Proof. The first two statements follow from (20)–(25). Woodbury expresses $\Delta \bar{P}$ through the $2q$ columns of $\bar{P} B_\Lambda$. For the gain,

$$\Delta \bar{K} = (\bar{P}^+ - \bar{P}) G^T + \bar{P}^+ V U^T. \quad (27)$$

The first term has column space contained in $\text{col}(\bar{P} B_\Lambda)$. Since V is a block of B_Λ ,

$$\bar{P}^+ V = \bar{P} V - \bar{P} B_\Lambda M_\Lambda^{-1} B_\Lambda^T \bar{P} V \quad (28)$$

also lies in $\text{col}(\bar{P} B_\Lambda)$. Hence $\text{col}(\Delta \bar{K}) \subseteq \text{col}(\bar{P} B_\Lambda)$ and the gain rank bound is $2q$, not the looser $3q$ bound obtained by adding ranks independently. $\square$

C. An Explicit Rank- $2q$ Gain Factorization

The gain-rank part of Theorem 2 can be made constructive. In information coordinates define

$$T = \bar{P} B_\Lambda, \quad E_V = \begin{bmatrix} 0_{q \times q} \\ I_q \end{bmatrix}, \quad (29)$$

so that $\bar{P} V = T E_V$. With $M_\Lambda = J_\Lambda^{-1} + B_\Lambda^T \bar{P} B_\Lambda$ and $\bar{K} = \bar{P} G^T$, direct substitution yields

$$\Delta \bar{K} = T Z_K, \quad (30)$$

where

$$Z_K = -M_\Lambda^{-1} B_\Lambda^T \bar{K} + E_V U^T \quad (31)$$

$$-M_\Lambda^{-1} B_\Lambda^T T E_V U^T. \quad (32)$$

Hence the gain correction can be stored/applied as an $n \times 2q$ panel times a $2q \times m$ panel. This is stronger than merely knowing that the rank is small: selected gain rows, state corrections, or covariance blocks can be updated without materializing the full dense gain difference.

D. State Correction With a Changing Right-Hand Side

Relinearization changes both the Gaussian matrix and the recentered residual (6). The rank recursion therefore does not imply that the new state is obtained by a rank update alone. The right-hand side must be reevaluated through $h(x_{j+1})$. However, once the new residual is known, the updated factor/inverse applies to it without a new dense factorization. In measurement coordinates this means computing $z_{j+1} = W_{j+1} \tilde{v}_{j+1}$ and then the state correction; in information coordinates the new normal-equation right-hand side is solved with $\bar{P}_{j+1}$. This separation prevents a common overclaim: the expensive *matrix* change is rank structured, while the nonlinear residual remains data dependent.

E. Log-Determinant, NIS, and Likelihood Without Dense Recomputations

The determinant lemma yields

$$\log |\bar{S}^+| = \log |\bar{S}| + \log |\det M_S|, \quad (33)$$

because $|\det J_S| = 1$. Let $z = W \tilde{v}$ and $g = B_S^T z$. Then (23) gives the explicit scalar recursion

$$\text{NIS}^+ = \tilde{v}^T W^+ \tilde{v} = \text{NIS} - g^T M_S^{-1} g. \quad (34)$$

Thus both terms of the Gaussian measurement log-likelihood

$$\ell = -\frac{1}{2} (\text{NIS} + \log |S| + m \log 2\pi) \quad (35)$$

can be updated through the same small core. In the safeguarded algorithm below, NIS is reported after an accepted update; it is not treated as an independent χ^2 draw inside repeated relinearizations of the same measurement.

Algorithm 1: Safeguarded rank-recursive IEKF

- 1) **Refresh**: evaluate exact H_0 , whiten to G_0 , and factor the chosen innovation/information matrix.
- 2) Form an IEKF candidate using the current Gaussian factors.
- 3) Backtrack on the true MAP objective until a decreasing step is found. If descent fails, return to Step 1.
- 4) Append $s_i = \xi_{i+1} - \xi_i$ and $y_i = \tilde{h}(\xi_{i+1}) - \tilde{h}(\xi_i)$. Form X, Y and $E = Y - GX$.
- 5) Compute a rank- q multisection correction (Euclidean or prior-metric equivalent in original coordinates) and factor it as UV^T .
- 6) Propagate the innovation/information factor through the rank- $2q$ update; do not refactor densely.
- 7) Stop on step/MAP-decrement tolerance; otherwise iterate.
- 8) Compute/report NIS, likelihood, and final posterior diagnostics after the accepted measurement update.

Fig. 1. The algorithm uses dense factorization only at refresh. The Gaussian algebra after each selected low-rank relinearization is exact.

F. Complexity With Factor-Formation Costs

The $\mathcal{O}(d^2q + q^3)$ Woodbury core alone understates total cost because the rank factors must also be formed. In dense arithmetic, operating in whitened coordinates requires at most

$$\boxed{\mathcal{O}(n^2q + mnq + d^2q + q^3)}, \quad (36)$$

where the terms correspond to prior-factor solves/whitening, GV or $G^T U$, the active-domain thin-panel update, and the $2q$ core. For $m \sim n \sim d$ and fixed small q , this is $\mathcal{O}(d^2q)$ versus $\mathcal{O}(d^3)$ dense refactorization. Sparse G , structured P^- , or already-whitened secants reduce the first two terms further. These counts are algorithmic operation models, not FPGA cycle or energy measurements.

V. SAFEGUARDED RANK-RECURSIVE IEKF

A. Low-Rank Surrogate, Exact Gaussian Step

The method intentionally separates approximation sources. The multisection correction is a low-rank surrogate for nonlinear curvature. Once G^+ is selected, the Gaussian update associated with that surrogate is exact through Section IV. This means a failure of the secant model should trigger a *Jacobian refresh*, not increasingly approximate matrix inversion.

At iteration j , let ξ_j be the normalized state. After evaluating $\tilde{h}(\xi_j)$, the current Gaussian factors produce a candidate IEKF/Gauss-Newton point $\hat{\xi}_{j+1}$. We use a monotone backtracking line search on (3),

$$\xi_{j+1} = \xi_j + \alpha_j(\hat{\xi}_{j+1} - \xi_j), \quad \alpha_j \in \{1, \frac{1}{2}, \frac{1}{4}, \dots\}, \quad (37)$$

until the MAP objective decreases. A new secant pair is then appended and the active q -column window is updated. If the secant Gram matrix is ill-conditioned or no acceptable descent step is found, an exact Jacobian refresh is performed. This is a standard globalization philosophy for quasi-Newton methods [7]; a new global-convergence theorem is not claimed here.

B. What the Method Does Not Assume

A tempting interpretation is that the exact Jacobian difference $H_{j+1} - H_j$ must itself have small numerical rank. That assumption is unnecessary and, in Section VII-D, empirically false for the AC IEEE 39-bus example. The method instead chooses a rank- q multisection *surrogate* that matches observed directional

curvature. This distinction is important: rank recursion is an exact consequence of the chosen model, not evidence that the exact nonlinear Hessian/Jacobian evolution is intrinsically low-rank.

C. Refresh Logic and Secant-Memory Adaptation

The role of q is deliberately modest. It is not increased because a statistical test “fails” on reused inner-loop data. Instead, q may be increased when the current curvature surrogate produces slow MAP decrease while the secant Gram matrix remains well conditioned. A practical deterministic policy is

$$q \in \{1, 2, 4\}, \quad q \leftarrow \min(2q, q_{\max}) \quad (38)$$

after N_s accepted but insufficiently productive steps, and an exact refresh when (i) the line search cannot find descent, (ii) $\text{cond}(X^T X)$ or $\text{cond}(X^T (P^-)^{-1} X)$ exceeds a threshold, or (iii) $q = q_{\max}$ and the MAP decrement stagnates. The current experiments use this refresh philosophy but do not tune it for optimal performance.

Because the measurement is reused across IEKF inner iterations, an inner-loop NIS value should not be interpreted as an independent χ^2 random variable. NIS and likelihood are nevertheless valuable outputs after the accepted measurement update and are algebraically cheap once the small core is available. This paper therefore separates numerical globalization (MAP descent) from post-update statistical diagnostics (NIS/likelihood).

VI. PROPOSED SQUARE-ROOT HARDWARE MAPPING

The explicit inverse formulas are useful analytically but should not be the long-term RTL state. Square-root and factored Kalman implementations are preferred under finite precision [18], [19]. Each update has the generic form

$$A^+ = A + BJB^T, \quad (39)$$

where J is only $2q \times 2q$ and may be indefinite. It can therefore be diagonalized locally and realized as a short sequence of rank-one factor updates/downdates, or mapped to a hyperbolic/J-orthogonal transformation as in robust Chandrasekhar filtering [11].

For $q_{\max} = 4$, the largest new dense core is 8×8 . The large kernels are regular matrix-thin-matrix products and triangular solves, well suited to DSP lanes and BRAM banking. The control path selects measurement or information domain based on m, n , sparsity, and factor availability. This is a *mapping proposal*, not a measured hardware contribution. VHDL, fixed-point analysis, synthesis, timing, resource use, and power are explicitly deferred to future work.

A. Planned VHDL Decomposition

A future synthesizable implementation can be partitioned into the modules in Table III. The table is intentionally phrased as a validation plan, not as completed hardware. The square-root factor bank and rank-event engine are the critical path for future RTL work; the exact-Jacobian refresh engine may

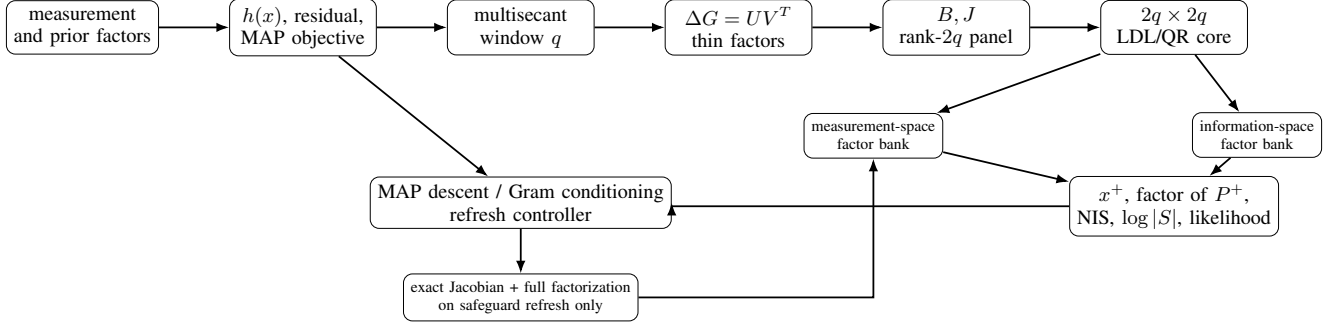


Fig. 2. Hardware mapping proposed for subsequent VHDL work. This paper does not report RTL synthesis. A square-root implementation would stream thin $d \times 2q$ panels through a factor bank and reserve a fully unrolled or time-multiplexed core for the small dense solve.

TABLE III
PLANNED VHDL BLOCKS; IMPLEMENTATION AND SYNTHESIS ARE PENDING.

Block	Main operation	Planned structure
Secant buffer	X, Y, E window	BRAM + small Gram engine
Prior solve	$L^{-1}s$ / metric product	triangular systolic/streaming
Thin-panel engine	$GV, G^T U$, factor products	DSP MAC lanes
Small core	$2q \times 2q$ LDL/QR solve	unrolled for $q \leq 4$
Factor update	signed rank-one updates/downdates	J-orthogonal / hyperbolic rotations
Refresh	exact Jacobian + full factorization	slower shared fallback
Controller	line search, conditioning, refresh	finite-state machine

be implemented on a slower shared datapath because it is not exercised on every inner iteration.

The first VHDL milestone should therefore be $q = 1$, for which the event core is 2×2 . A second configuration can support $q = 2$ and $q = 4$ through time multiplexing. The evaluation protocol should report bit width, factor-drift tests, cycles per rank event, cycles per refresh, Fmax, LUT/FF/DSP/BRAM, and energy per accepted IEKF update. None of these quantities is estimated in the present paper.

VII. COST LAWS AND DESIGN REGIMES

A. Amortization Across IEKF Inner Iterations

The rank-recursive formulation is most useful when a measurement update requires multiple relinearizations. Let C_D denote the cost of one dense Jacobian-dependent Gaussian rebuild/refactorization and $C_R(q)$ the cost of one rank- q event after the initial factorization. If an exact IEKF uses k inner linearizations, its algebraic cost is approximately

$$C_{\text{exact}} \approx kC_D. \quad (40)$$

With no intermediate refresh, the rank-recursive cost is

$$C_{\text{RR}} \approx C_D + (k-1)C_R(q). \quad (41)$$

Defining $\rho(q) = C_R(q)/C_D$, the idealized gain is

$$G_k(q) = \frac{k}{1 + (k-1)\rho(q)}. \quad (42)$$

This expression makes two limits explicit. If $k = 1$, there is no within-update factorization to reuse and the method offers no benefit. If $\rho \ll 1$ and k is moderate, the gain approaches k . Thus the method targets nonlinear measurements that genuinely require repeated relinearization, rather than ordinary one-shot EKF updates.

Suppose r_f exact refreshes occur after the initial factorization. Then

$$C_{\text{RR}} \approx (1 + r_f)C_D + (k-1-r_f)C_R(q), \quad (43)$$

which immediately yields a refresh-rate break-even condition. For a given measured ρ , a hardware designer can determine how often dense refresh may occur before rank recursion loses its advantage. This is more informative than quoting a single FLOP ratio from one matrix size.

B. Measurement Versus Information Domain

The rank event itself is dual. Ignoring common formation terms, a measurement-space update scales roughly as m^2q while an information-space update scales as n^2q . A first-order selector is therefore

$$\text{domain} = \arg \min\{\hat{C}_m, \hat{C}_n\}, \quad (44)$$

with

$$\hat{C}_m = c_1 mnq + c_2 m^2 q + c_3 q^3, \quad (45)$$

$$\hat{C}_n = c_1 mnq + c_2 n^2 q + c_3 q^3. \quad (46)$$

The constants depend on whether G or G^T is streamed naturally, whether P^- is dense, and which square-root factor is already resident in memory. The rule “choose $\min(m, n)$ ” is therefore only a dimensional heuristic; a real implementation should choose the lower measured streaming/factor cost.

C. When a Larger Secant Memory Helps

The event rank grows linearly with q , but nonlinear iterations may decrease. Let $k(q)$ be the number of accepted IEKF steps. The relevant optimization is

$$q^* = \arg \min_q [C_D + (k(q)-1)C_R(q)], \quad (47)$$

not $\arg \min_q k(q)$. This observation motivates a small discrete set such as $q \in \{1, 2, 4\}$ instead of an unconstrained memory.

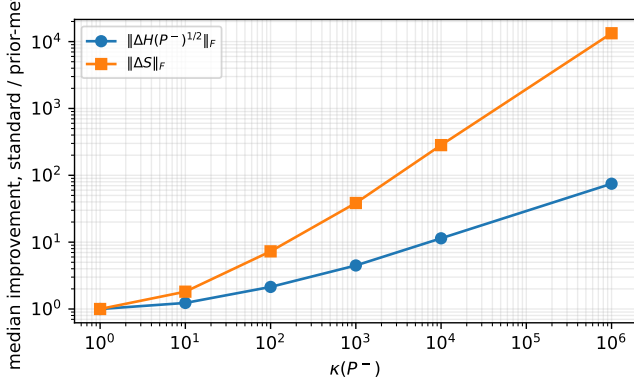


Fig. 3. Effect of prior anisotropy in 3000 secant trials. Both updates satisfy the same secant equations; the prior-metric rule minimizes the statistically weighted least-change criterion and the bound in (16).

The AC experiment later shows that $q = 2$ reduces mean iteration count relative to $q = 1$ for both tested secant metrics, but the improvement is modest; a hardware-optimal q will therefore depend on the measured $C_R(q)$ curve.

D. Interaction With the Time Update

The current theory addresses low-rank change *within* one iterated measurement update, where P^- is fixed. Between measurement times, the process model changes the prior and can destroy the previously factored innovation matrix. Classical Chandrasekhar recursions exploit low-rank Riccati differences across time [10], [11]; combining those temporal displacement-rank ideas with the present relinearization-rank event is promising but outside the present scope. Consequently, the guaranteed reuse window here is the IEKF inner loop, with one dense initialization per measurement update unless additional temporal structure is available.

VIII. EXPERIMENTS

All experiments use double precision to isolate the algebraic and nonlinear effects. Scripts, seeds, and CSV outputs accompany the manuscript. Hardware speedup is not inferred from wall-clock Python time.

A. A. Prior-Metric Secant Geometry

We first isolate Theorem 1. For random 24-state/18-measurement problems, 500 secants are generated at each $\text{cond}(P^-) \in \{1, 10, 10^2, 10^3, 10^4, 10^6\}$, for 3000 trials total. Euclidean Broyden and (12) satisfy the same secant constraint to machine precision. Figure 3 reports the ratio “Euclidean/prior-metric” for the weighted Jacobian change and the induced innovation-covariance perturbation. The prior metric is identical when P^- is isotropic, but becomes increasingly different as the prior becomes anisotropic. At $\text{cond}(P^-) = 10^2, 10^3, 10^4$, the median $\|\Delta S\|_F$ reduction is $7.29\times$, $38.5\times$, and $282.8\times$, respectively. These are stress-test geometry results, not claims that realistic filters always possess such condition numbers or that smaller ΔS guarantees fewer nonlinear iterations.

TABLE IV
RANK-RECURSION EXACTNESS AND ARITHMETIC PROXY ($n = 48, m = 64$, 40 TRIALS PER q).

q	max rel. S^{-1}	max rel. K	max $ \Delta \log S $	dense/rank proxy
1	6.9×10^{-16}	1.2×10^{-15}	4.3×10^{-14}	19.35
2	6.6×10^{-16}	1.2×10^{-15}	3.6×10^{-14}	9.57
4	6.6×10^{-16}	1.2×10^{-15}	4.3×10^{-14}	4.69
8	6.5×10^{-16}	1.2×10^{-15}	4.3×10^{-14}	2.24

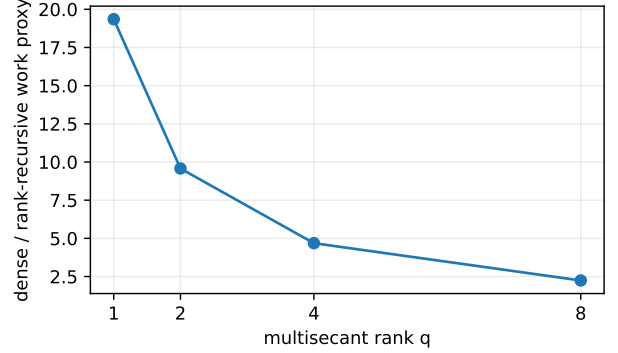


Fig. 4. Representative dense/rank-recursive arithmetic proxy versus multisecant rank. The advantage vanishes as q approaches the ambient dimension, motivating small adaptive memories.

B. B. Algebraic Exactness for $q = 1, 2, 4, 8$

For each $q \in \{1, 2, 4, 8\}$, 40 random problems with $n = 48, m = 64$ are updated using a rank- q Jacobian change. Dense recomputation of S^{-1} , the information inverse, gain, log-determinant, and NIS is compared with the rank-recursive formulas. Table IV shows that all differences remain at floating-point roundoff. The work proxy in the final column counts dominant dense products after U, V are available and is included only to show the scaling trend; it is fully specified in Appendix B and is not a cycle-level hardware claim.

C. C. Full AC IEEE 39-Bus Rank-Recursive IEKF

The main nonlinear experiment uses the solved MATPOWER IEEE 39-bus case [20]. The state contains 38 non-reference voltage angles plus all 39 voltage magnitudes, hence $n = 77$. Sixteen PMU locations provide direct angle and magnitude measurements; active/reactive branch-flow measurements are included at PMU incident line ends using the full complex branch model with resistance, reactance, line charging, and transformer tap ratios. This yields $m = 108$ whitened measurements. The true state is a coherent perturbation around the solved MATPOWER operating point, and 12 independent noise/prior realizations are evaluated for each $\text{cond}(P^-) \in \{10^2, 10^3, 10^4\}$.

The exact IEKF evaluates a finite-difference Jacobian at every inner iteration. The rank-recursive methods evaluate it initially and only again on safeguard refresh. Both Euclidean and prior-metric multisecants are tested with $q = 1, 2$; Table V reports $q = 2$, which is a useful compromise in this case. Crucially, the estimator itself propagates S^{-1} using (23); the

dense inverse is formed only in the reference implementation and in evaluation-only consistency checks.

Evaluation uses four metrics: exact Jacobian evaluations, inner iterations, posterior Mahalanobis distance

$$d_P = \sqrt{(\hat{x} - x_{\text{IEKF}})^T (P_{\text{IEKF}}^+)^{-1} (\hat{x} - x_{\text{IEKF}})}, \quad (48)$$

and signed MAP objective gap $\Phi(\hat{x}) - \Phi(x_{\text{IEKF}})$. This replaces the ambiguous “cost gap” and avoids relying on coordinate-dependent Euclidean distance.

Two conclusions are more important than claiming a winner between secant metrics. First, both rank-recursive quasi-Newton variants reduce exact-Jacobian evaluations from roughly 4–4.5 to about 1–1.25 while remaining close to the exact IEKF. Second, the prior-metric geometry does *not* uniformly dominate Euclidean Broyden in nonlinear convergence: it is better at $\kappa(P^-) = 10^2$ in this experiment, comparable in iteration count at 10^4 , but produces a larger posterior deviation there. This negative result narrows the claim appropriately: Theorem 1 controls the least-change geometry, not the global nonlinear trajectory.

The rank recursion itself is exact within numerical precision throughout the AC runs. Across all tested updates, the relative identity error in the structured S update is below 1.1×10^{-15} and the recursive inverse differs from dense recomputation by at most 1.0×10^{-11} , the latter reflecting accumulated floating-point roundoff over multiple sequential updates.

D. D. Effective Rank of the Exact Jacobian Change

For each exact-IEKF trajectory we also compute singular values of

$$\Delta H_{\text{exact}}(P^-)^{1/2}. \quad (49)$$

The median rank needed to capture 99% of Frobenius energy is 29.5, 24.5, and 21 for $\text{cond}(P^-) = 10^2, 10^3, 10^4$; for 99.9% energy it is 45.5, 40, and 33. Thus the physical test does not support a claim that exact relinearization is intrinsically rank one or rank two. The computational opportunity comes instead from *limited-memory directional curvature modeling*: a small multisecant panel can be useful even when the exact Jacobian difference is not globally low-rank.

E. E. Effect of Multisecant Memory

Aggregating all 36 AC trials, increasing the memory from $q = 1$ to $q = 2$ reduces mean inner iterations from 8.11 to 7.39 for Euclidean multisecant Broyden and from 8.39 to 7.36 for the prior-metric variant. Exact-Jacobian evaluation counts remain near one per measurement update because refreshes are uncommon. The posterior-distance distributions change only slightly: the median d_P values are 0.0288 and 0.0282 for Euclidean $q = 1, 2$, and 0.0328 and 0.0323 for prior-metric $q = 1, 2$. In this test, $q = 2$ therefore buys modest nonlinear progress without materially changing statistical fidelity. Larger memory is not assumed to be monotonically beneficial; it increases the event core and may overfit secants spanning a region in which the Jacobian changes nonlinearly.

This ablation supports an adaptive rather than fixed large-memory design. The relevant trade is not only “more curvature is better,” but

$$\text{nonlinear iterations saved} \quad \text{vs.} \quad \text{rank-event cost } \mathcal{O}(qd^2). \quad (50)$$

A hardware implementation should choose q using end-to-end latency/energy rather than iteration count alone.

IX. EXTENSION: A UNIFIED LOW-RANK EVENT FORM

The same arithmetic primitive can absorb low-rank changes that are not caused by relinearization. This is useful for interpreting the method relative to Chandrasekhar filtering and for designing a reusable implementation block.

Proposition 2 (combined low-rank event). Suppose, in original coordinates,

$$H^+ = H + UV^T, \quad \text{rank}(UV^T) \leq q, \quad (51)$$

$$P^+ = P + L_P C_P L_P^T, \quad \text{rank}(L_P C_P L_P^T) \leq r, \quad (52)$$

$$R^+ = R + L_R C_R L_R^T, \quad \text{rank}(L_R C_R L_R^T) \leq s, \quad (53)$$

where C_P, C_R may be signed small matrices. Then the innovation matrix $S = HPH^T + R$ satisfies

$$\boxed{\text{rank}(S^+ - S) \leq 2q + r + s.} \quad (54)$$

Proof. Add and subtract $H^+PH^{+T} + R$:

$$S^+ - S = \underbrace{(H^+PH^{+T} - HPH^T)}_{\text{rank} \leq 2q} \quad (55)$$

$$+ \underbrace{H^+L_P C_P L_P^T H^{+T}}_{\text{rank} \leq r} + \underbrace{L_R C_R L_R^T}_{\text{rank} \leq s}. \quad (56)$$

The rank of a sum is no larger than the sum of ranks. $\square$

Equation (54) does not replace specialized fast filters. Rather, it identifies a common interface: any small event can be passed to a generic signed panel update BJB^T . A Jacobian relinearization contributes up to $2q$ columns; a low-rank prior change contributes r ; and sensor/noise reconfiguration contributes s . Classical Chandrasekhar methods provide more specialized temporal recursions for particular Riccati differences [10], [11]; the proposition only shows that a future hardware block need not be exclusive to the IEKF inner loop.

A. Structured and Tensor-Generated Secant Panels

The rank-recursive Gaussian algebra depends on the panel width q , not on how the factors U, V were obtained. Consequently, a structured nonlinear model may generate ΔH from sparse local updates, Kronecker factors, CP/Tucker factors, or tensor-train contractions without changing the $2q$ Gaussian core. This is the useful point of contact with tensor methods: a tensor representation is valuable if it produces a small set of matrix outer-product directions, not because the entire Kalman filter must be rewritten as a tensor network. Such extensions are not evaluated here and are intentionally kept outside the novelty claims.

TABLE V

FULL AC IEEE 39-BUS EXPERIMENT, $q = 2$ (12 TRIALS PER PRIOR CONDITION NUMBER). “JAC.” COUNTS ALGORITHMIC EXACT-JACOBIAN EVALUATIONS; THE TERMINAL JACOBIAN USED ONLY TO SCORE POSTERIOR DISTANCE IS EXCLUDED.

$\kappa(P^-)$	Method	mean iter.	mean Jac.	mean refresh	median d_P	P95 d_P	median MAP gap
10^2	Euclidean	8.33	1.00	0.00	0.0218	0.0314	2.37×10^{-4}
	Prior-metric	7.75	1.17	0.17	0.0167	0.0261	1.41×10^{-4}
10^3	Euclidean	6.92	1.17	0.17	0.0321	0.0434	5.16×10^{-4}
	Prior-metric	7.50	1.08	0.08	0.0362	0.0579	6.56×10^{-4}
10^4	Euclidean	6.92	1.25	0.25	0.0330	0.0540	5.45×10^{-4}
	Prior-metric	6.83	1.00	0.00	0.0429	0.1660	9.20×10^{-4}
10^2	Exact IEKF	3.00	4.00	—	0	0	0
10^3	Exact IEKF	3.17	4.17	—	0	0	0
10^4	Exact IEKF	3.50	4.50	—	0	0	0

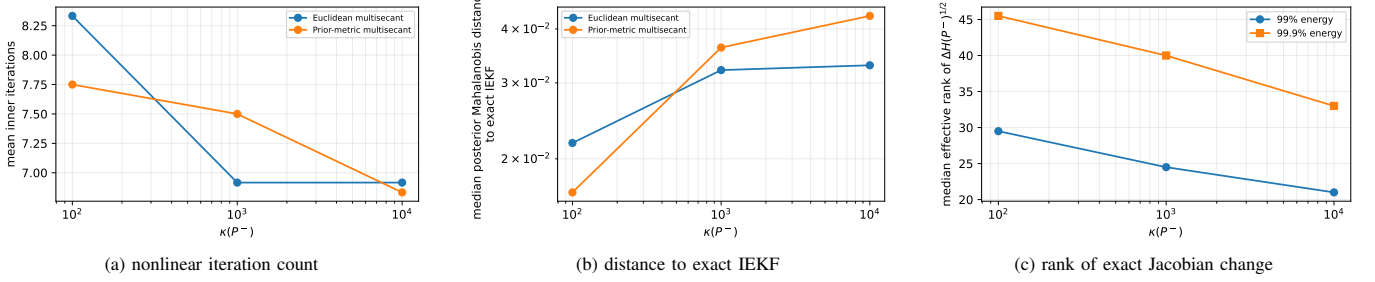


Fig. 5. AC IEEE 39-bus results. The prior-metric rule is not universally superior in nonlinear convergence. Panel (c) is equally important: exact Jacobian changes require median ranks 21–30 for 99% energy, so the proposed small- q method should be understood as a multisecant surrogate, not a low-rank truncation of the exact Jacobian.

B. Sensor Addition, Dropout, and Noise Adaptation

A change in a small number of measurement-noise variances is a low-rank modification of R . Likewise, adding or removing a small set of channels can often be represented as a bordered or signed low-rank information update after reindexing. Proposition 2 therefore suggests a deterministic fast path for occasional sensor events. The safeguard philosophy remains the same: if the event rank exceeds the configured small-core capacity, the implementation falls back to full factorization rather than forcing an inaccurate compression.

X. DISCUSSION AND LIMITATIONS

A. What Is New and What Is Reused

The ingredients—Broyden/multisecant updates, weighted least-change norms, Woodbury identities, incremental factorizations, and square-root Kalman methods—are established. The contribution is their filtering-specific closure: a deliberately low-rank IEKF relinearization generates a rank- $2q$ event in both dual Gaussian domains, allowing several posterior quantities to be propagated exactly through the same small core. This is narrower than claiming a new quasi-Newton method and broader than applying Woodbury to one matrix inverse.

B. No Hardware Result Yet

The proposed mapping in Fig. 2 is intentionally not called an implemented FPGA architecture. No VHDL has yet been synthesized; no Fmax, latency, DSP, LUT, BRAM, power, or fixed-point result is reported. The next hardware study must compare a square-root rank-recursive implementation against

a conventional Cholesky IEKF and relevant approximate-hardware baselines such as KalmMind [16], [17]. The key hardware question is whether the reduced serial core and thin-panel streaming outweigh factor-bank traffic and refresh cost.

C. Numerical Stability and Refresh

Explicit inverse recursions can accumulate roundoff, as suggested by the 10^{-11} worst-case inverse discrepancy in the sequential AC experiment even though the underlying rank identity is accurate to 10^{-15} . A production implementation should therefore maintain square-root factors and use update/downdate transformations. Exact refreshes also reset numerical drift. Finite-precision guarantees remain open and should be treated as a separate hardware-validation problem.

D. Convergence Scope

The safeguarded algorithm uses MAP descent and exact refresh to avoid blindly propagating a poor secant model. We do not claim a new global convergence theorem. Establishing local superlinear behavior or Dennis–Moré-type conditions for the prior-metric multisecant under periodic refresh is an attractive analytical extension, but is not required for the algebraic rank-closure results proved here.

E. Finite-Precision Implications

The AC experiment provides an instructive numerical separation. The structured matrix identity $S^+ = S + BJB^T$ is satisfied to about 10^{-15} relative error, while repeated explicit inverse recursion drifts to roughly 10^{-11} . This is not a failure of the rank theorem; it is the familiar sensitivity of recursively

updated inverses. A factor implementation should therefore periodically compare inexpensive residuals such as

$$\frac{\|I - S\widehat{W}\|_F}{\sqrt{m}} \quad (57)$$

against a threshold or, preferably, avoid storing W explicitly altogether. In square-root form the signed $2q$ event can be processed as a sequence of rank-one updates/downdates after a local eigendecomposition/LDL factorization of J . The finite-precision VHDL study should report both posterior error and factor residual over long refresh intervals, because a design can be algebraically exact in real arithmetic yet numerically unacceptable at aggressive fixed-point widths.

F. A Useful Failure Mode

If the multisecant Gram matrix becomes ill-conditioned, the nominal rank q overstates the number of independent curvature directions. Rather than regularizing silently and changing the secant problem, Algorithm 1 drops the oldest secants until the Gram matrix is usable; persistent failure triggers a full refresh. This policy preserves a clear interpretation of q as an effective independent-memory rank. Regularized or soft-secanted variants may be valuable, especially when measurements are noisy, but then Theorem 1 must be modified because the exact hard constraint $\Delta HX = E$ no longer holds and the left measurement metric need not cancel.

XI. CONCLUSION

This paper reformulates a specific computational bottleneck in iterated Kalman filtering: once a nonlinear relinearization is represented by a rank- q multisecant correction, the Gaussian update should not be treated as an unrelated dense problem. The induced innovation, information, posterior-covariance, and gain changes all have rank at most $2q$, and the associated inverse, log-determinant, NIS, likelihood, and posterior quantities admit exact small-core recursions. A prior-metric least-change specialization provides coordinate-covariant secant geometry and can strongly reduce innovation perturbation under anisotropic priors, although full AC experiments show that it does not universally outperform Euclidean Broyden in nonlinear convergence.

The resulting picture is therefore precise: low-rank relinearization is a *curvature model*, not an assertion that the exact Jacobian change is low-rank; Gaussian propagation is exact for that model; and dense Jacobian/factor refresh remains the safeguard. The full AC IEEE 39-bus experiment reduces analytic Jacobian evaluations by roughly a factor of three to four while maintaining small posterior deviations and machine-precision rank identities. The next step is deliberately hardware-specific: implement the square-root rank- $2q$ update in synthesizable VHDL, quantify fixed-point drift, and measure post-route latency, resources, and energy. Those results are left pending rather than inferred from arithmetic counts.

APPENDIX A

PROOF DETAILS AND EXPLICIT RECURSIONS

A. Prior-Metric Least-Change Proof

With $Z = R^{-1/2}\Delta H(P^-)^{1/2}$, $A = (P^-)^{-1/2}X$, and $B = R^{-1/2}E$, problem (11) becomes $\min_Z \|Z\|_F$ subject to

$ZA = B$. For full-column-rank A , the unique minimum-norm solution is $Z^* = B(A^T A)^{-1}A^T$. Substitution gives (12). This also makes explicit why the left measurement metric cancels: $R^{1/2}B = E$. If secants are softened rather than enforced exactly, or regularization is added in measurement coordinates, R need not cancel; that is a different optimization problem.

B. Posterior and Gain Update in Information Coordinates

Let $\bar{P} = \bar{\Lambda}^{-1}$ and

$$M_\Lambda = J_\Lambda^{-1} + B_\Lambda^T \bar{P} B_\Lambda. \quad (58)$$

Then

$$\bar{P}^+ = \bar{P} - \bar{P} B_\Lambda M_\Lambda^{-1} B_\Lambda^T \bar{P}. \quad (59)$$

Thus both posterior covariance and gain changes can be represented by thin factors derived from $\bar{P} B_\Lambda$. One need not materialize the full difference matrix to update selected covariance blocks or state corrections.

C. Explicit NIS and Log-Determinant Update

Equation (34) requires only the old innovation solve $z = W\tilde{r}$, the thin projection $g = B_S^T z$, and the already-factorized M_S . Likewise (33) adds the log absolute determinant of the same small core. These formulas explain why statistical monitoring does not require an additional dense factorization once the rank event has been processed.

D. Affine-Covariance Proof

Under $x = Tz$, $P_x = T P_z T^T$, $X_x = T X_z$, and $H_z = H_x T$. Then

$$\Delta H_x T = E(X_x^T P_x^{-1} X_x)^{-1} X_x^T P_x^{-1} T \quad (60)$$

$$= E(X_z^T P_z^{-1} X_z)^{-1} X_z^T P_z^{-1} = \Delta H_z, \quad (61)$$

which proves the stated coordinate covariance. Ordinary Euclidean Broyden has this property only for orthogonal/scalar coordinate changes because its metric $X^T X$ is not transformed with the estimator uncertainty.

E. Rank-Closure Proof for the Posterior Covariance

From (25), Woodbury gives

$$\Delta \bar{P} = -\bar{P} B_\Lambda M_\Lambda^{-1} B_\Lambda^T \bar{P}. \quad (62)$$

The difference factors through the $2q$ columns of $\bar{P} B_\Lambda$, immediately proving $\text{rank}(\Delta \bar{P}) \leq 2q$. The same factor appears in (30), explaining why covariance and gain corrections share a common thin state-space basis. This shared basis is attractive for future hardware because one panel can feed multiple downstream quantities.

APPENDIX B

RECONSTRUCTABLE OPERATION-COUNT MODEL

For a representative measurement-space update with n states, m measurements, and rank q , after the secant factors U, V are available, the dominant dense rank-recursive kernels are approximately

$$PV : n^2q, \quad (63)$$

$$H(PV) : mnq, \quad (64)$$

$$WB_S : 2m^2q, \quad (65)$$

$$B_S^T WB_S : 4mq^2, \quad (66)$$

$$(WB_S)M_S^{-1}(WB_S)^T : 4m^2q, \quad (67)$$

$$M_S \text{ factor/solve} : \mathcal{O}((2q)^3). \quad (68)$$

The information-domain expressions follow by interchanging the active dimension. By contrast, a dense innovation route requires formation of $HP^{-1}H^T$, factorization of an $m \times m$ SPD matrix, and gain solves; an implementation can associate the products differently depending on m, n and structure. For the synthetic $n = 48, m = 64$ study in Table IV, the reported proxy uses

$$C_{\text{rank}} = 6m^2q + 4mq^2 + \frac{(2q)^3}{3}, \quad (69)$$

and

$$C_{\text{dense}} = \frac{m^3}{3} + 2m^2n, \quad (70)$$

with common Jacobian/model evaluation excluded. This proxy is deliberately simple and reproducible; it does not include cache/BRAM traffic, parallel lane count, or critical path, and therefore must not be interpreted as measured speedup.

APPENDIX C

REPRODUCIBILITY NOTES

The AC experiment uses MATPOWER case39 solved voltages and branch parameters [20]. The state is $[\theta_{\text{ref}}, V]$ with bus 31 as the reference angle. PMU buses are $\{2, 6, 10, 14, 16, 20, 23, 25, 29, 30, 32, 33, 35, 36, 38, 39\}$. Angle and magnitude standard deviations are 8×10^{-4} , while active/reactive line-flow standard deviations are 10^{-2} p.u.; measurements are whitened before filtering. The supplied scripts generate all Monte Carlo tables, the exact-Jacobian effective-rank diagnostic, and the rank-recursion identity checks from fixed seeds.

A. Full AC Measurement Equations

For reproducibility, each branch (i, j) with series admittance $y_{ij} = 1/(r_{ij} + jx_{ij})$, total charging susceptance b_{ij} , and off-nominal tap t_{ij} uses the standard MATPOWER primitive

$$Y_{ff} = (y_{ij} + jb_{ij}/2)/t_{ij}^2, \quad Y_{ft} = -y_{ij}/t_{ij}, \quad (71)$$

$$Y_{tf} = -y_{ij}/t_{ij}, \quad Y_{tt} = y_{ij} + jb_{ij}/2. \quad (72)$$

With complex bus voltages $\mathcal{V}_i = V_i e^{j\theta_i}$,

$$I_{ij} = Y_{ff}\mathcal{V}_i + Y_{ft}\mathcal{V}_j, \quad S_{ij} = \mathcal{V}_i I_{ij}^*. \quad (73)$$

The measurement function includes direct PMU-bus V_i, θ_i and $\Re(S_{ij}), \Im(S_{ij})$ at PMU incident line ends. This experiment is

therefore nonlinear in both voltage magnitude and angle and includes branch resistance, line charging, and transformer tap ratios; it is no longer the fixed-magnitude sinusoidal angle-only example used in the preliminary manuscript.

The true test state is a coherent perturbation around the solved case39 operating point. Angle perturbations are generated from a reduced network Laplacian excited at several stressed buses and scaled to a maximum 0.10 rad displacement; a small spatially diffused voltage-magnitude perturbation is superimposed. This produces a controlled nonlinear estimation test, not a time-domain electromechanical simulation. The distinction is retained in the claims.

B. Monte Carlo Sample Size and Interpretation

The full AC study uses 12 independent trials per prior condition number, which is sufficient to expose qualitative differences but not to support fine-grained statistical superiority claims. Accordingly, the paper reports medians/P95 values and avoids significance language. The much larger 3000-trial experiment in Section VII-A is used only for the algebraic secant-geometry claim. A future application paper should expand the AC Monte Carlo campaign, add dynamic process trajectories/topology events, and report confidence intervals and failure probabilities.

REFERENCES

- [1] B. M. Bell and F. W. Cathey, "The iterated kalman filter update as a gauss-newton method," *IEEE Transactions on Automatic Control*, vol. 38, no. 2, pp. 294–297, 1993.
- [2] H. Auvinen, J. M. Bardsley, H. Haario, and T. Kauranne, "Large-scale kalman filtering using the limited memory bfgs method," *Electronic Transactions on Numerical Analysis*, vol. 35, pp. 217–233, 2009.
- [3] T. Mukherjee, D. Varshney, K. K. Kottakki, and M. Bhushan, "Broyden's update based extended kalman filter for nonlinear state estimation," *Journal of Process Control*, vol. 105, pp. 267–282, 2021.
- [4] A. Kullberg, M. A. Skoglund, I. Skog, and G. Hendeby, "On the relationship between iterated statistical linearization and quasi-newton methods," *IEEE Signal Processing Letters*, vol. 30, pp. 1777–1781, 2023.
- [5] C. G. Broyden, "A class of methods for solving nonlinear simultaneous equations," *Mathematics of Computation*, vol. 19, no. 92, pp. 577–593, 1965.
- [6] J. E. Dennis and H. Wolkowicz, "Sizing and least-change secant methods," *SIAM Journal on Numerical Analysis*, vol. 30, no. 5, pp. 1291–1314, 1993.
- [7] J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. Springer, 2006.
- [8] M. Kaess, A. Ranganathan, and F. Dellaert, "isam: Incremental smoothing and mapping," *IEEE Transactions on Robotics*, vol. 24, no. 6, pp. 1365–1378, 2008.
- [9] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. Leonard, and F. Dellaert, "isam2: Incremental smoothing and mapping using the bayes tree," *The International Journal of Robotics Research*, vol. 31, no. 2, pp. 216–235, 2012.
- [10] M. Morf, G. S. Sidhu, and T. Kailath, "Some new algorithms for recursive estimation in constant, linear, discrete-time systems," *IEEE Transactions on Automatic Control*, vol. 19, no. 4, pp. 315–323, 1974.
- [11] M. V. Kulikova, "Hyperbolic svd-based kalman filtering for chandrasekhar recursion," *IET Control Theory & Applications*, vol. 13, pp. 1525–1531, 2019.
- [12] Y. Liu, C.-S. Bouganis, and P. Y. K. Cheung, "Efficient mapping of a kalman filter into an fpga using taylor expansion," in *2007 International Conference on Field Programmable Logic and Applications*, 2007, pp. 490–495.
- [13] K. S. Babu and K. P. Detroja, "Inverse free kalman filter using approximate inverse of diagonally dominant matrices," *IEEE Control Systems Letters*, vol. 3, no. 1, pp. 120–125, 2019.

- [14] J. M. Bardsley, A. Parker, A. Solonen, and M. Howard, "Krylov space approximate kalman filtering," *Numerical Linear Algebra with Applications*, vol. 20, no. 2, pp. 171–184, 2013.
- [15] M. Pförtner, J. Wenger, J. Cockayne, and P. Hennig, "Computation-aware kalman filtering and smoothing," in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 258, 2025, pp. 2071–2079.
- [16] G. Eichler, J. Zuckerman, and L. P. Carloni, "Kalmmind: A configurable kalman filter design framework for embedded brain–computer interfaces," in *2025 Design, Automation & Test in Europe Conference*, 2025, pp. 1–2.
- [17] —, "An energy-efficient kalman filter architecture with tunable accuracy for brain–computer interfaces," in *2025 62nd ACM/IEEE Design Automation Conference*, 2025.
- [18] G. J. Bierman, *Factorization Methods for Discrete Sequential Estimation*. Academic Press, 1977.
- [19] M. V. Kulikova and G. Y. Kulikov, "Svd-based square-root kalman filtering: A survey of existing implementation methods and novel techniques," *Digital Signal Processing*, vol. 166, p. 105391, 2025.
- [20] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, "Matpower: Steady-state operations, planning, and analysis tools for power systems research and education," *IEEE Transactions on Power Systems*, vol. 26, no. 1, pp. 12–19, 2011.